QuantForge

Final Comprehensive Backtest Report

2026-04-29

11 Symbols x 10Y x 4 Modes (Baseline / Regime / Risk / LLM)

Watchlist: GOOGL AAPL MSFT NVDA SPY QQQ

Non-Watchlist: AMZN TSLA JPM JNJ XOM

Strategy Architecture

1. Strategy Overview (ce lue jia gou)

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ben bao gao ce shi QuantForge wan zheng guan xian,
han gai 11 zhi biao de, 10 nian hui ce, 4 zhong mo shi:

Mode A (Baseline):  All rule signals, no filter.
Mode B (+ Regime):  Skip BEAR/CRISIS via VIX+ADX+MA60.
Mode C (+ Risk):    B + QuantScanner + PositionSizer +
                    CircuitBreaker (15% DD halt).
Mode D (+ LLM):     B + BacktestLLMFilter (Claude via
                    AMD Gateway). De-identifies data.
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2. Integrated Modules

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RegimeDetector     VIX>30=CRISIS, ADX+MA60 for trend
QuantScanner       TechnicalFactor(45%)+Cross(20%)+Sent(35%)
PositionSizer      5% base x score x regime discount
CircuitBreaker     DD>15% halt 22d, 8 consecutive loss halt
BacktestLLMFilter  De-identified prompt to Claude,
                   confidence>=55 to enter, retry+backoff
CostModel          RT ~1.09% (comm+slip+impact)
WalkForward        Rolling window, 70% pass rate
MonteCarlo         1000 shuffles for tail risk
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3. Test Universe

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Watchlist (self-selected):
  GOOGL  AAPL  MSFT  NVDA  SPY  QQQ

Non-Watchlist (diverse sectors):
  AMZN (e-commerce/cloud)  TSLA (EV, high vol)
  JPM (banking)  JNJ (healthcare, defensive)
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XOM (energy/oil)
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4. Entry / Exit Rules

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Entry: MA20/60 crossover up + RSI oversold bounce
Stop:  Entry - 2x ATR(14)
Target: Entry + 3x ATR(14)
Time:  60 trading day max hold
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5. Anti-Look-Ahead (LLM)

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Symbol -> STOCK_A, dates -> Day 0/-1/-2,
prices -> % changes, only indicators sent.
LLM cannot identify stock or time period.
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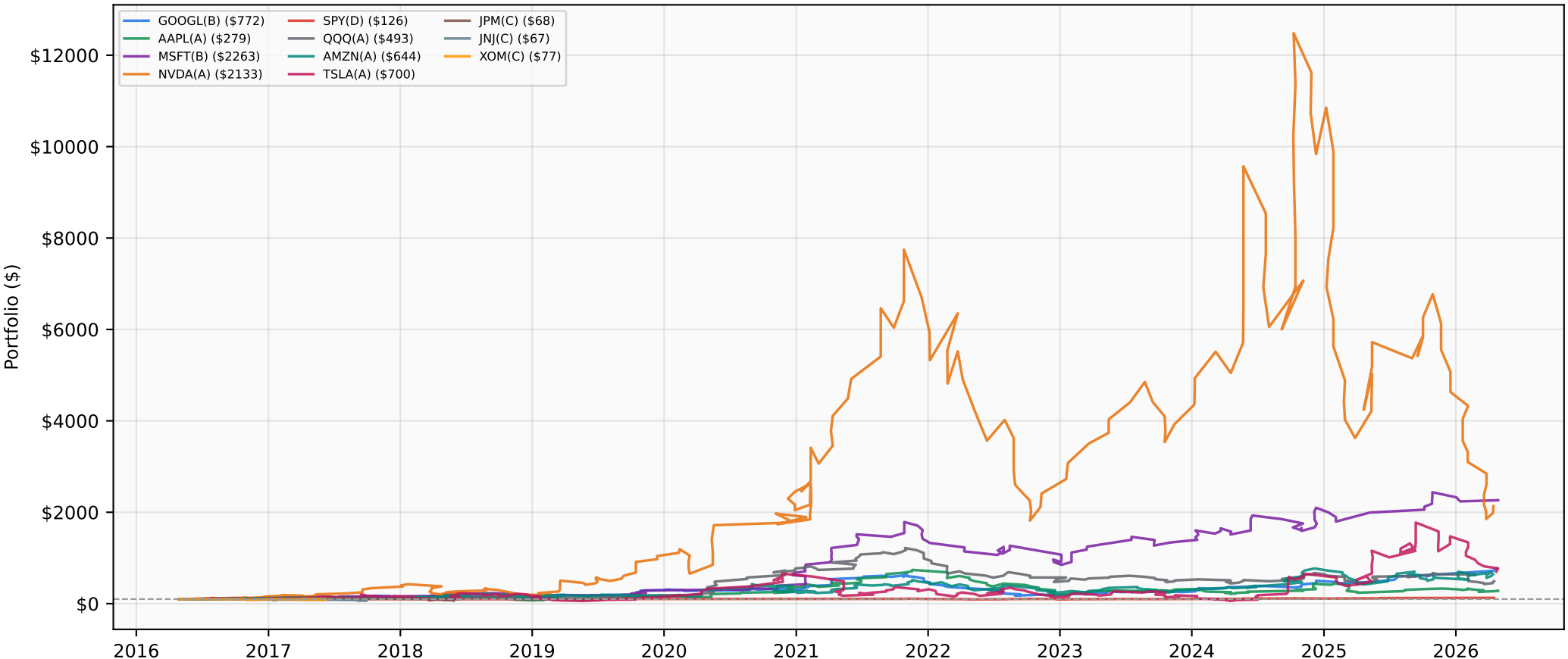
Watchlist Results (GOOGL/AAPL/MSFT/NVDA/SPY/QQQ)

Symbol	Group	Mode	Trades	Win%	Total%	Ann%	Sharpe	MDD%	PF	$100->	Verdict
GOOGL	Watchlist	A	220	60.5	+668.4	+22.7	2.68	86.4	1.47	$768	VALID
GOOGL	Watchlist	B	165	64.8	+672.1	+22.8	3.22	73.0	1.69	$772	VALID
GOOGL	Watchlist	C	92	67.4	+241.0	+13.1	2.77	46.6	1.85	$98	VALID
GOOGL	Watchlist	D	36	61.1	+16.9	+1.6	0.67	35.5	1.26	$117	MARGINAL
AAPL	Watchlist	A	192	59.4	+178.8	+10.8	1.62	74.1	1.28	$279	VALID
AAPL	Watchlist	B	138	60.1	+90.2	+6.7	1.29	60.8	1.26	$190	VALID
AAPL	Watchlist	C	117	60.7	+62.3	+5.0	1.10	55.5	1.24	$101	VALID
AAPL	Watchlist	D	38	65.8	+53.7	+4.4	1.48	28.2	1.64	$154	VALID
MSFT	Watchlist	A	260	62.7	+1199.2	+29.3	3.06	82.5	1.53	$1299	VALID
MSFT	Watchlist	B	210	68.6	+2163.0	+36.7	4.65	52.5	1.95	$2263	VALID
MSFT	Watchlist	C	177	66.1	+813.2	+24.8	3.60	45.6	1.76	$103	VALID
MSFT	Watchlist	D	44	40.9	-41.5	-5.2	-1.25	48.7	0.68	$58	REJECT
NVDA	Watchlist	A	223	57.0	+2033.0	+35.9	2.75	85.2	1.49	$2133	VALID
NVDA	Watchlist	B	163	57.7	+659.5	+22.6	2.35	86.0	1.47	$760	VALID
NVDA	Watchlist	C	124	62.9	+1073.0	+28.0	3.02	57.4	1.76	$104	VALID
NVDA	Watchlist	D	44	56.8	+84.8	+6.3	1.36	28.0	1.53	$185	VALID
SPY	Watchlist	A	220	52.3	-28.0	-3.2	-0.17	56.6	0.97	$72	REJECT
SPY	Watchlist	B	165	52.1	-46.6	-6.1	-1.24	52.6	0.81	$53	REJECT
SPY	Watchlist	C	10	10.0	-21.1	-2.4	-3.95	21.2	0.12	$100	INSUFFICIENT
SPY	Watchlist	D	30	70.0	+26.4	+2.4	1.32	13.8	1.66	$126	VALID
QQQ	Watchlist	A	212	58.5	+392.6	+17.3	2.41	65.9	1.47	$493	VALID
QQQ	Watchlist	B	158	58.9	+76.3	+5.8	1.34	51.2	1.26	$176	VALID
QQQ	Watchlist	C	128	58.6	+39.1	+3.4	0.95	35.1	1.19	$101	MARGINAL
QQQ	Watchlist	D	36	66.7	+51.7	+4.3	1.64	22.5	1.75	$152	VALID

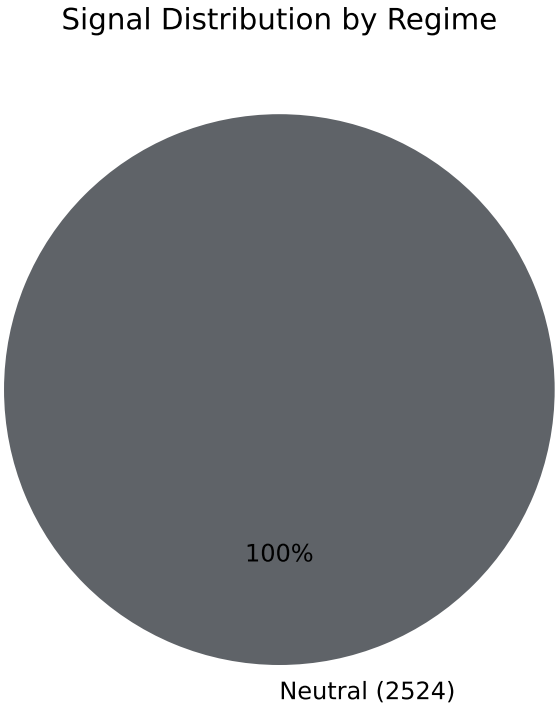
Non-Watchlist Results (AMZN/TSLA/JPM/JNJ/XOM)

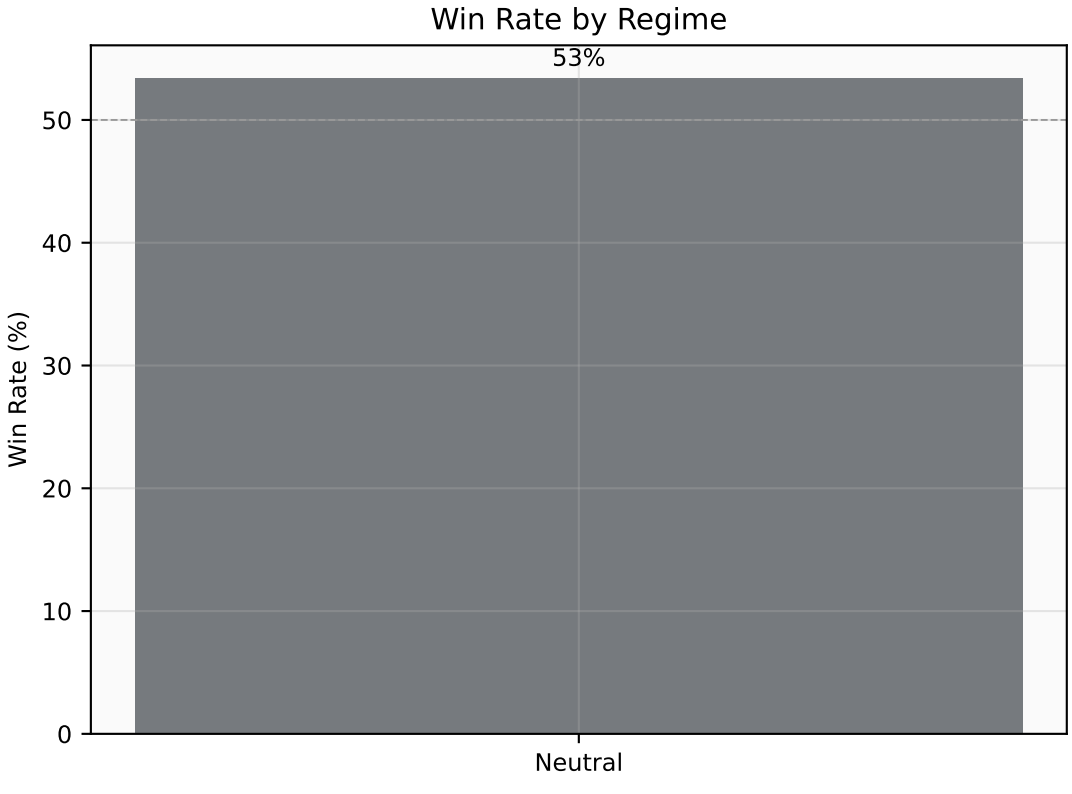
Symbol	Group	Mode	Trades	Win%	Total%	Ann%	Sharpe	MDD%	PF	$100->	Verdict
AMZN	Non-WL	A	236	56.8	+544.5	+20.5	2.24	55.0	1.38	$644	VALID
AMZN	Non-WL	B	175	59.4	+476.0	+19.2	2.56	38.0	1.52	$576	VALID
AMZN	Non-WL	C	147	60.5	+439.2	+18.4	2.64	31.3	1.60	$103	VALID
AMZN	Non-WL	D	48	56.2	+34.9	+3.0	0.90	34.5	1.32	$135	MARGINAL
TSLA	Non-WL	A	241	49.4	+599.6	+21.5	1.94	90.7	1.31	$700	MARGINAL
TSLA	Non-WL	B	172	48.8	+57.6	+4.7	1.02	93.2	1.18	$158	MARGINAL
TSLA	Non-WL	C	37	43.2	-22.7	-2.5	-0.32	60.5	0.90	$100	REJECT
TSLA	Non-WL	D	52	46.2	-5.9	-0.6	0.28	58.5	1.08	$94	REJECT
JPM	Non-WL	A	229	49.3	-14.4	-1.6	0.27	84.0	1.04	$86	REJECT
JPM	Non-WL	B	180	47.2	-56.2	-8.0	-0.86	86.5	0.88	$44	REJECT
JPM	Non-WL	C	22	36.4	-31.9	-3.8	-1.63	41.5	0.49	$100	INSUFFICIENT
JPM	Non-WL	D	43	37.2	-43.8	-5.6	-1.67	44.7	0.59	$56	REJECT
JNJ	Non-WL	A	233	41.2	-92.3	-22.7	-4.03	92.7	0.58	$8	REJECT
JNJ	Non-WL	B	164	42.7	-80.2	-15.0	-3.20	81.5	0.60	$20	REJECT
JNJ	Non-WL	C	23	30.4	-33.0	-3.9	-3.50	35.6	0.24	$101	INSUFFICIENT
JNJ	Non-WL	D	40	47.5	-26.3	-3.0	-1.22	32.4	0.67	$74	REJECT
XOM	Non-WL	A	258	42.6	-70.3	-11.4	-0.71	87.8	0.90	$30	REJECT
XOM	Non-WL	B	186	41.4	-79.0	-14.5	-1.84	91.0	0.75	$21	REJECT
XOM	Non-WL	C	20	30.0	-23.2	-2.6	-1.71	27.1	0.47	$105	INSUFFICIENT
XOM	Non-WL	D	36	44.4	-13.6	-1.4	-0.32	30.3	0.89	$86	REJECT

Best Mode per Symbol -- $100 Equity

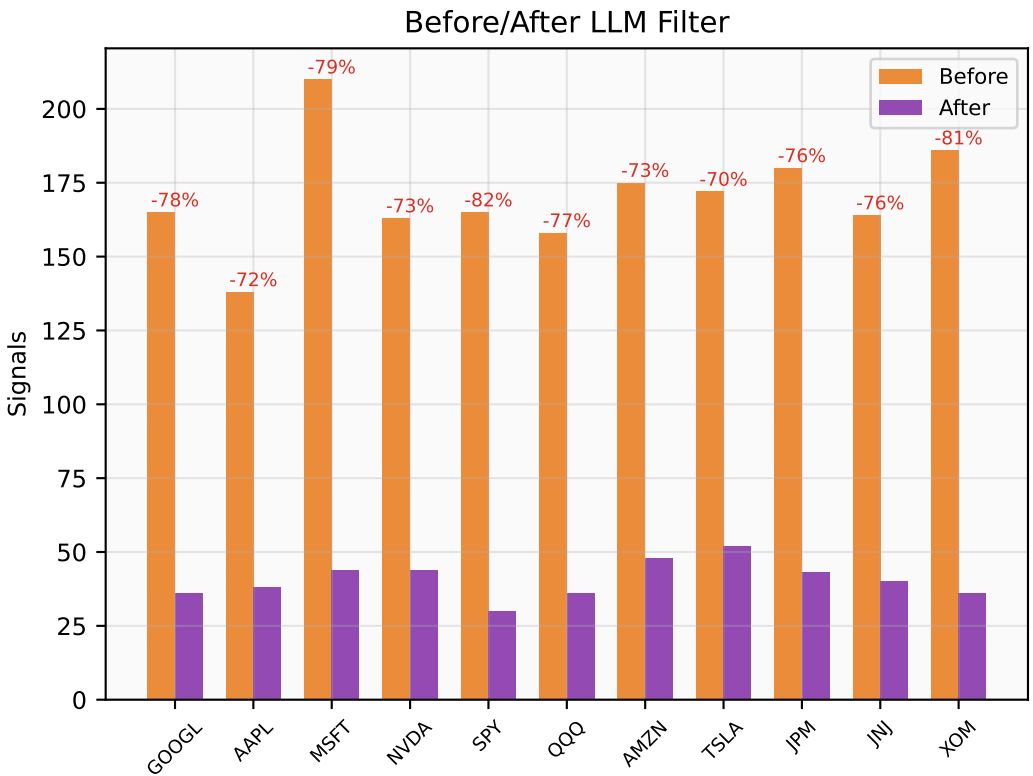


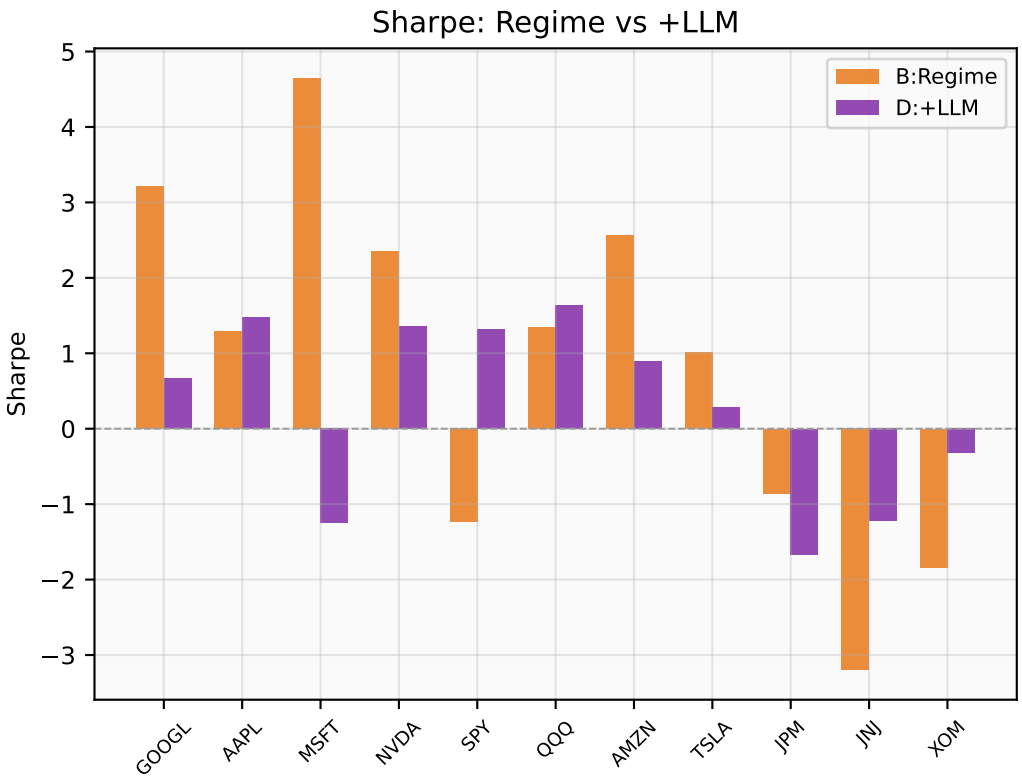
Regime Distribution & Win Rate



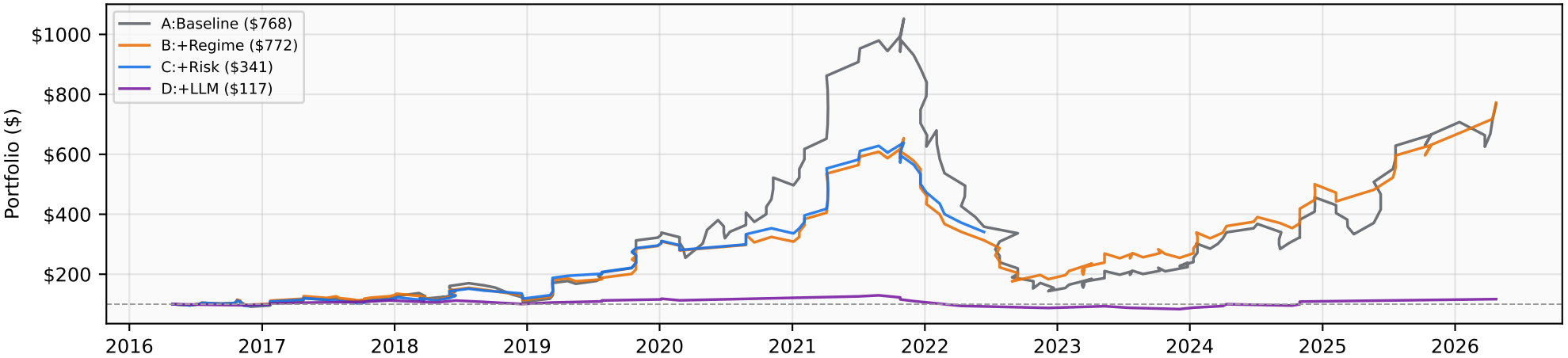


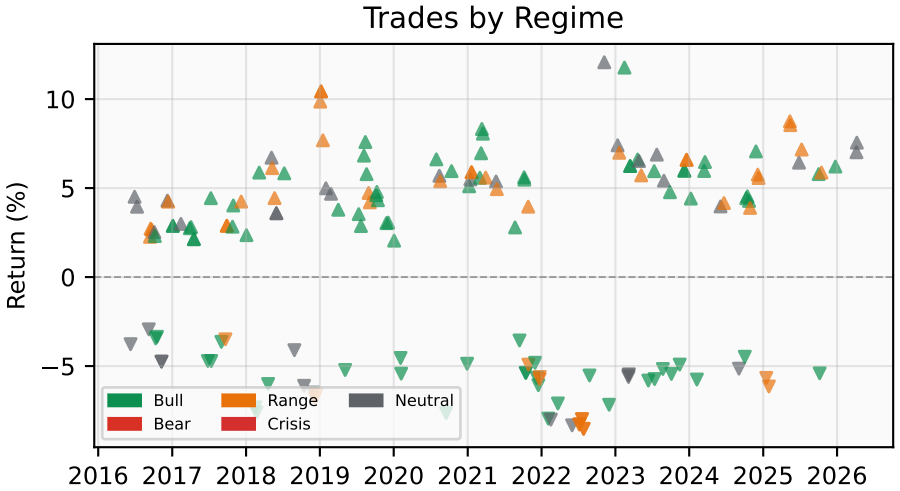
LLM Filter (Mode D) Impact

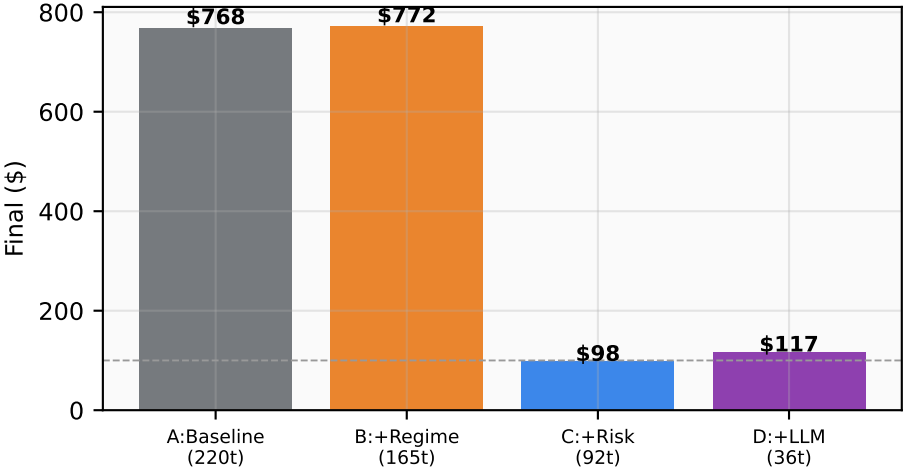




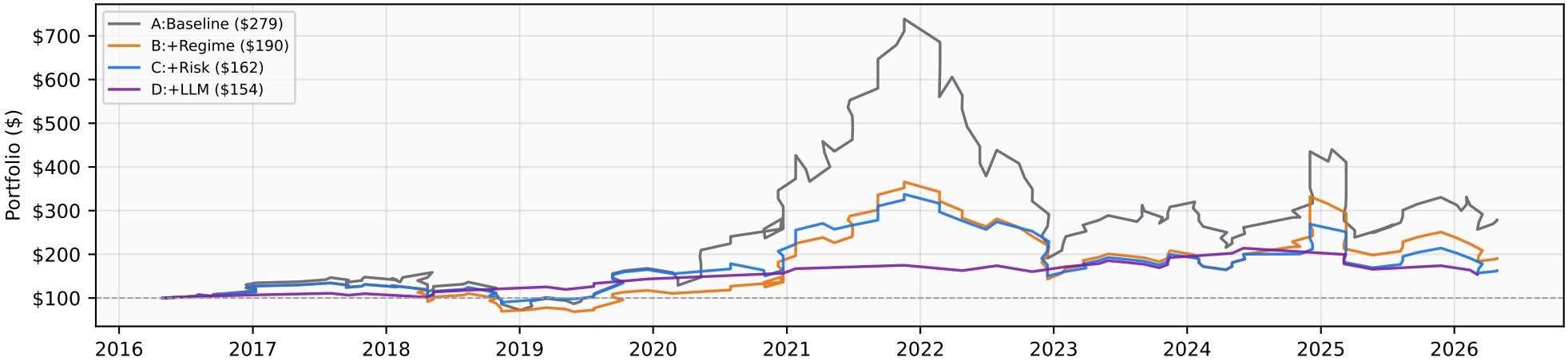
GOOGL -- 4-Mode Comparison ($100)

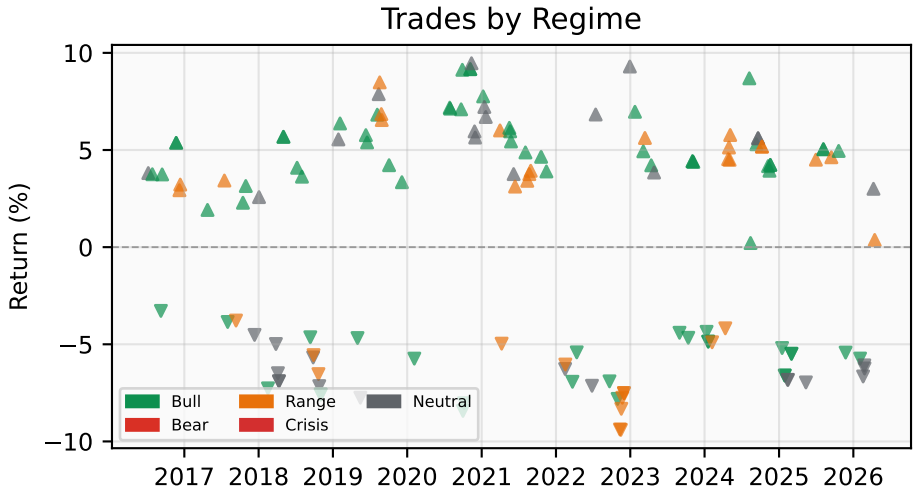


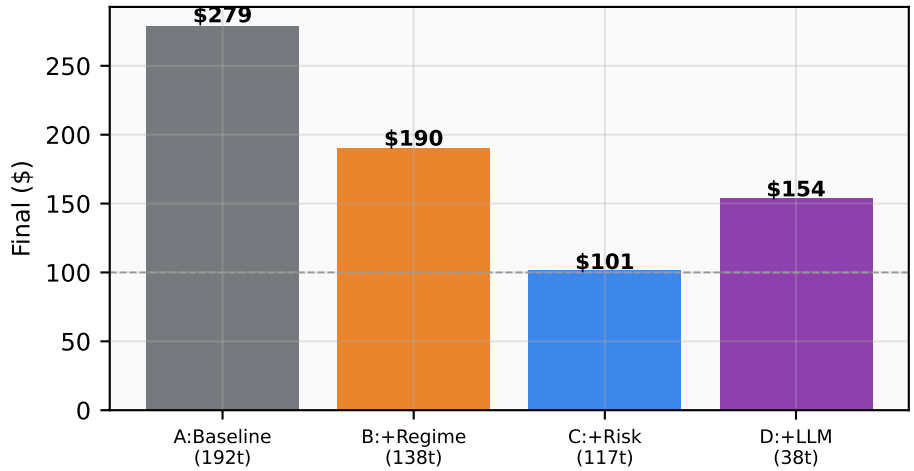




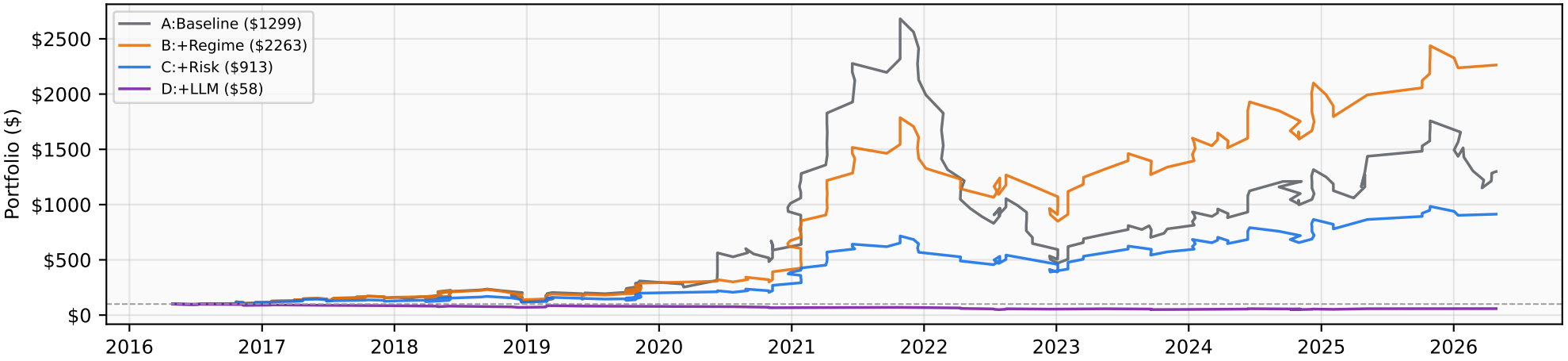
AAPL -- 4-Mode Comparison ($100)

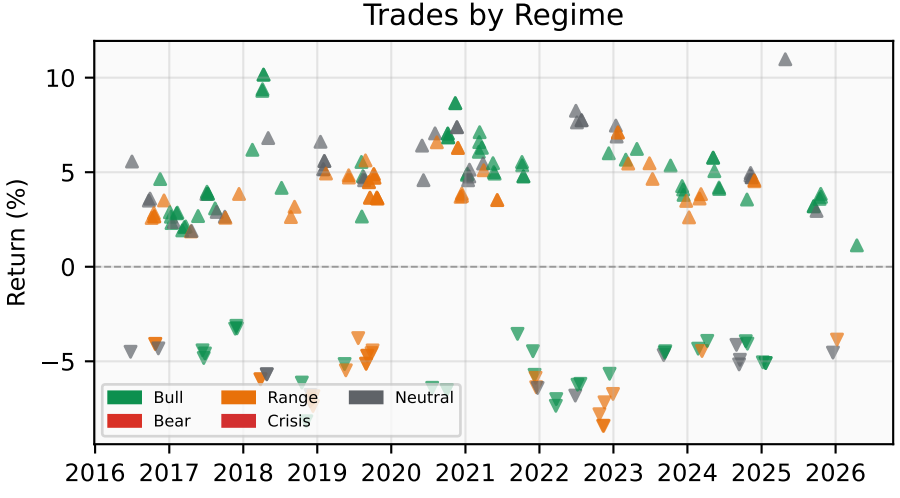


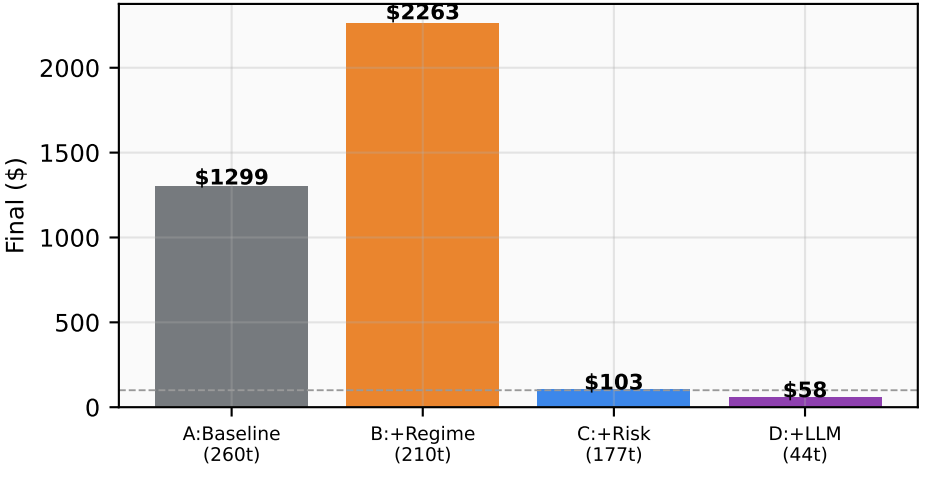




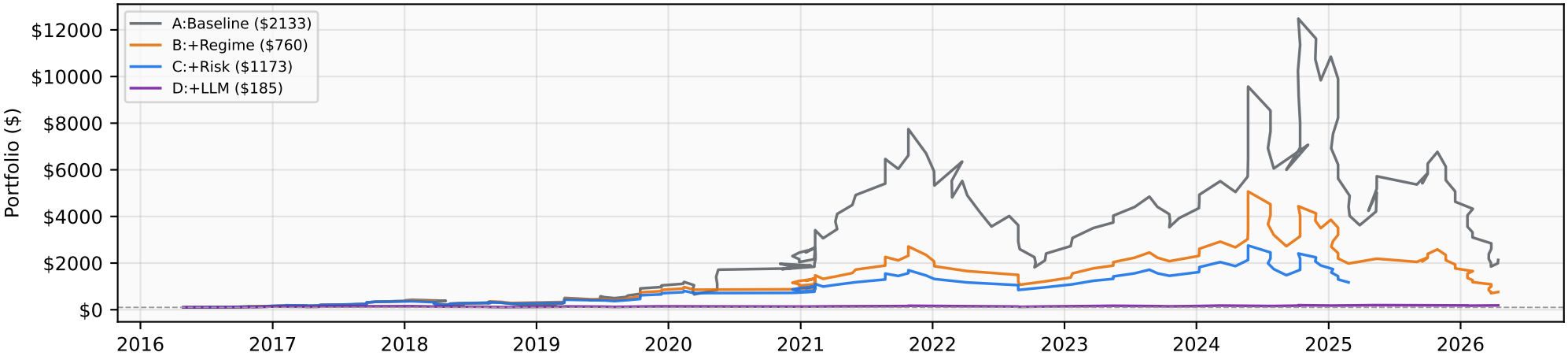
MSFT -- 4-Mode Comparison ($100)

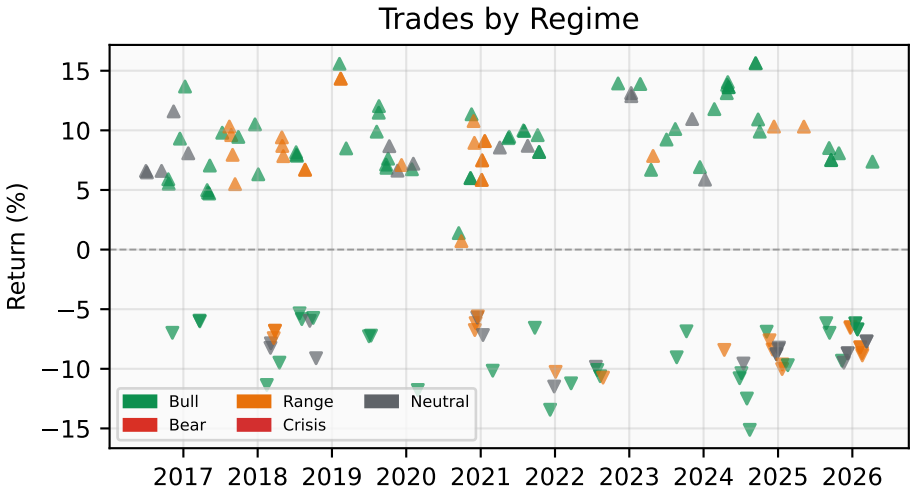


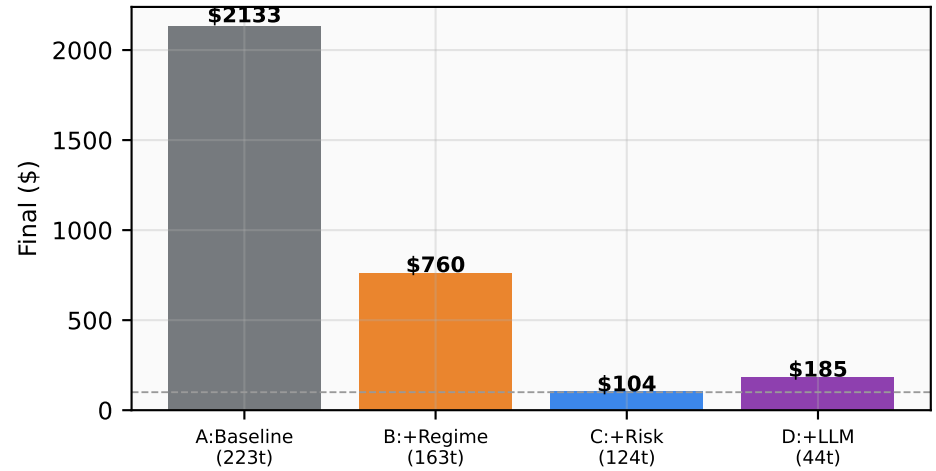




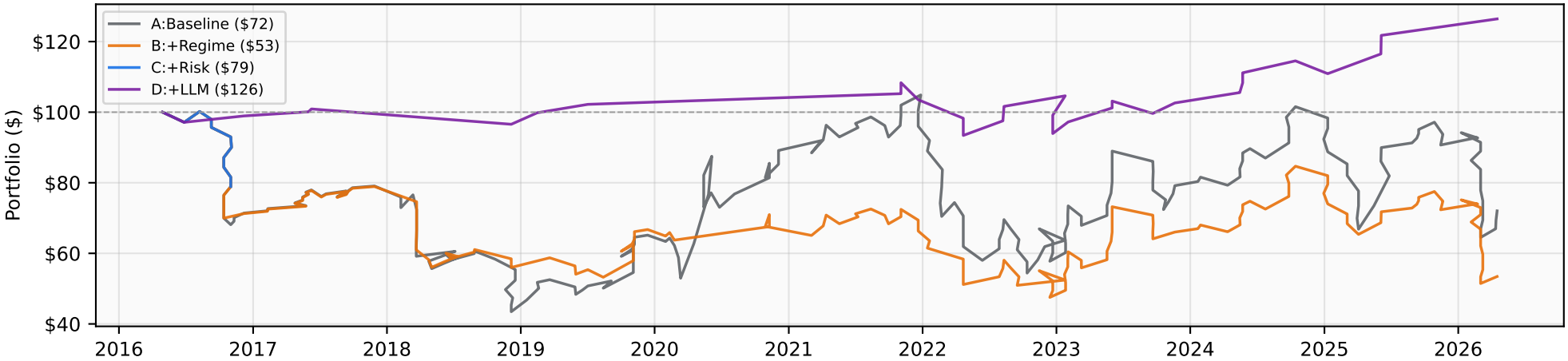
NVDA -- 4-Mode Comparison ($100)

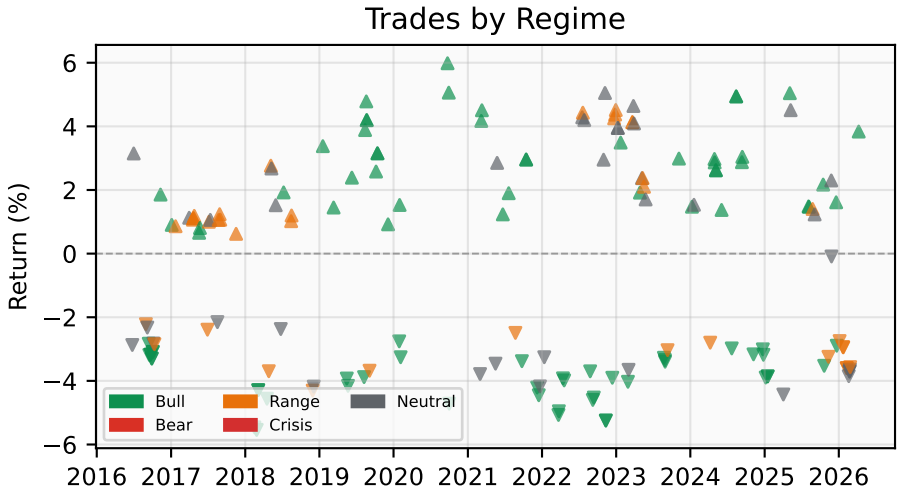


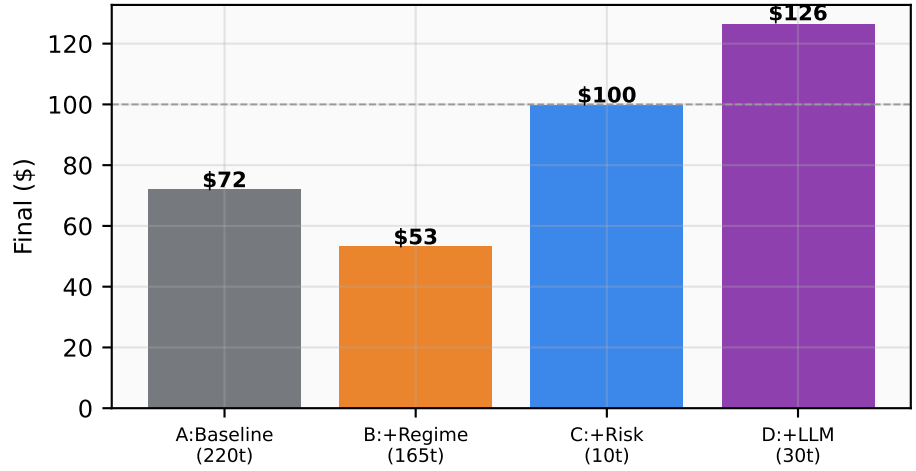




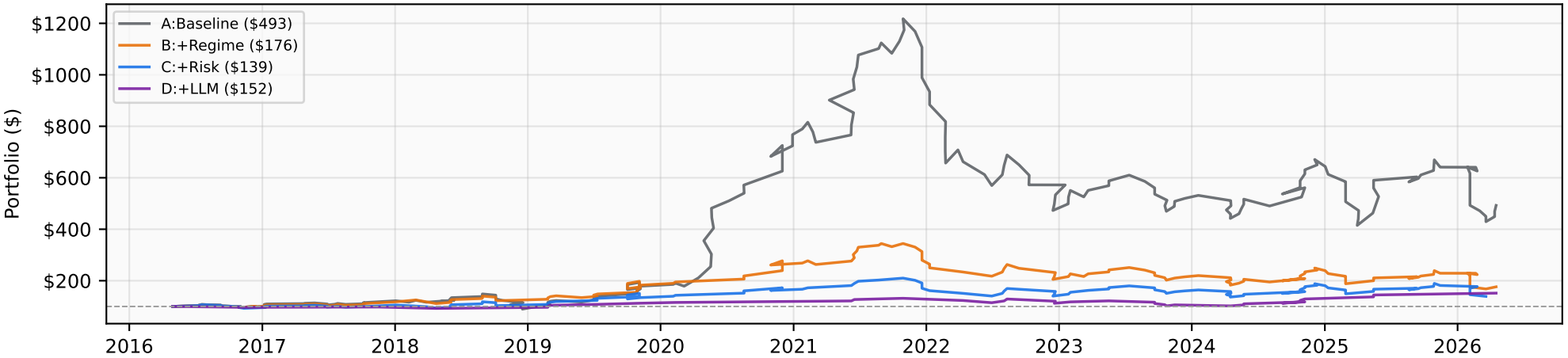
SPY -- 4-Mode Comparison ($100)

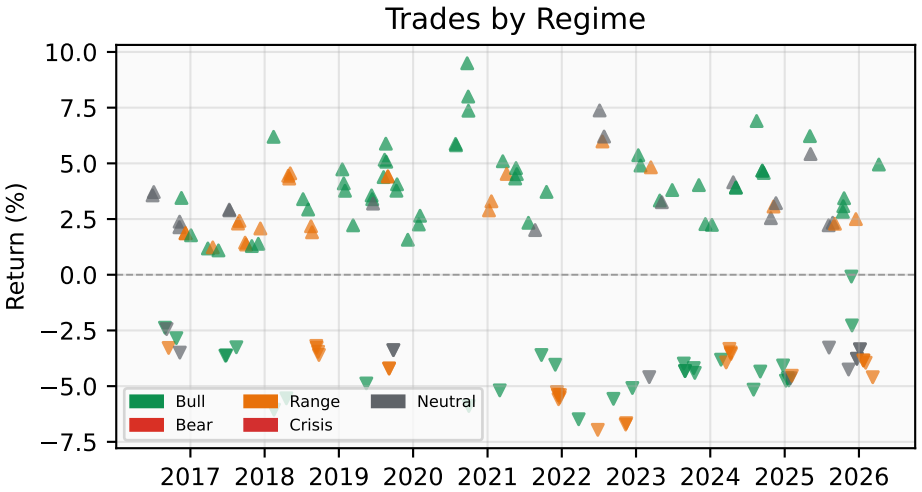


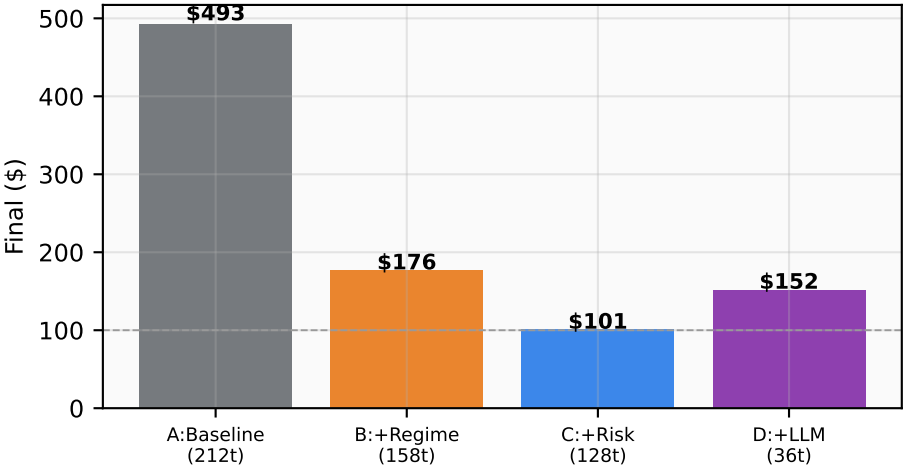




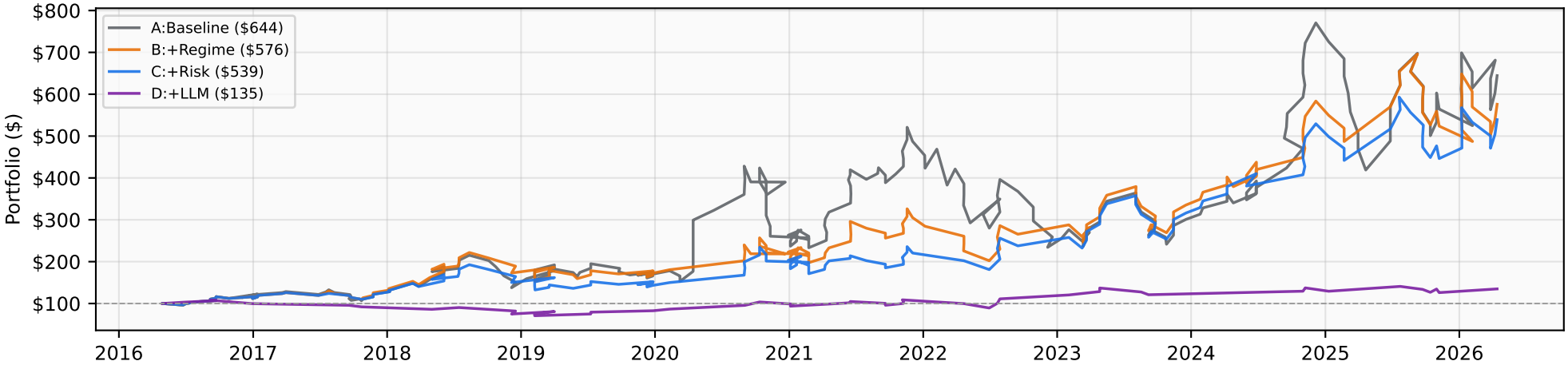
QQQ -- 4-Mode Comparison ($100)

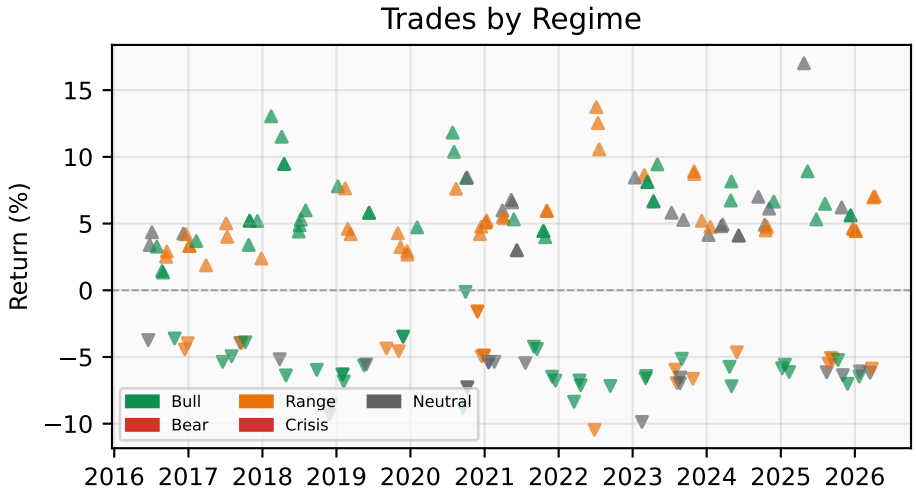


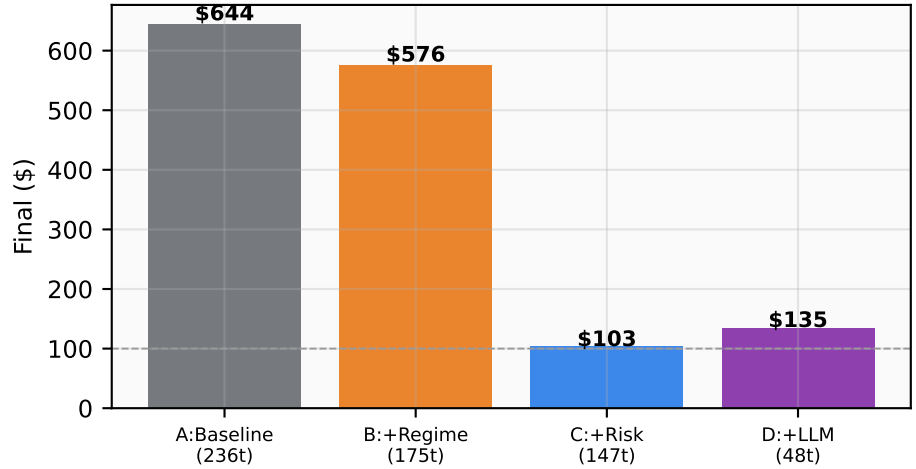




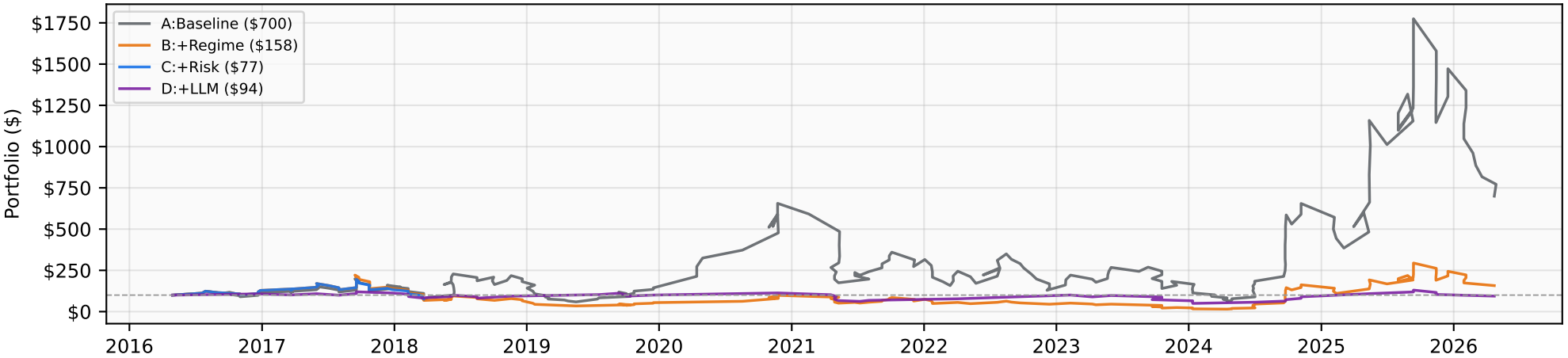
AMZN -- 4-Mode Comparison ($100)

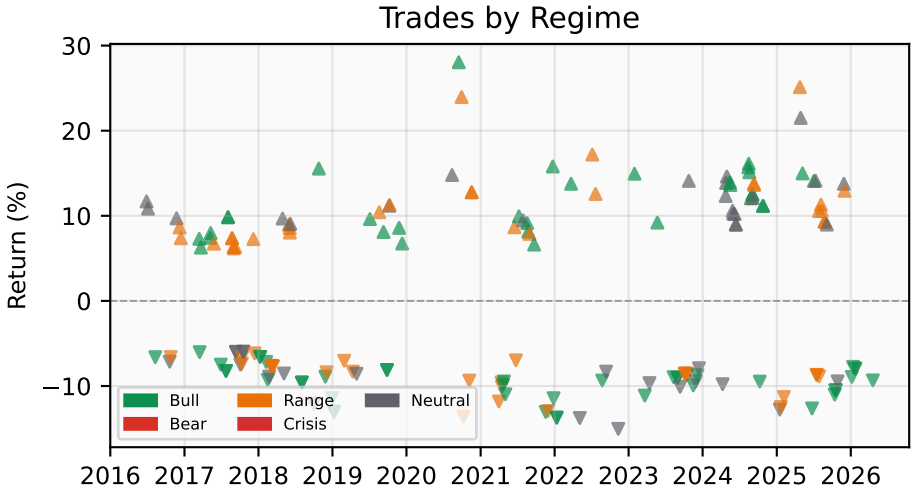


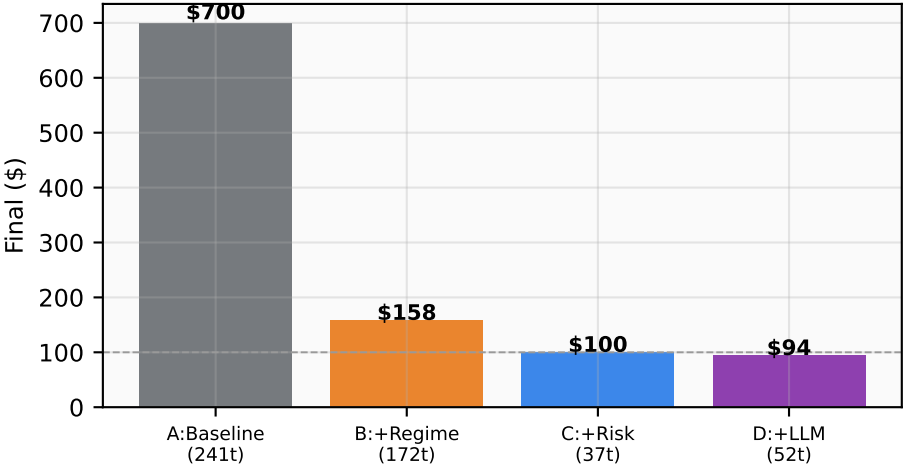




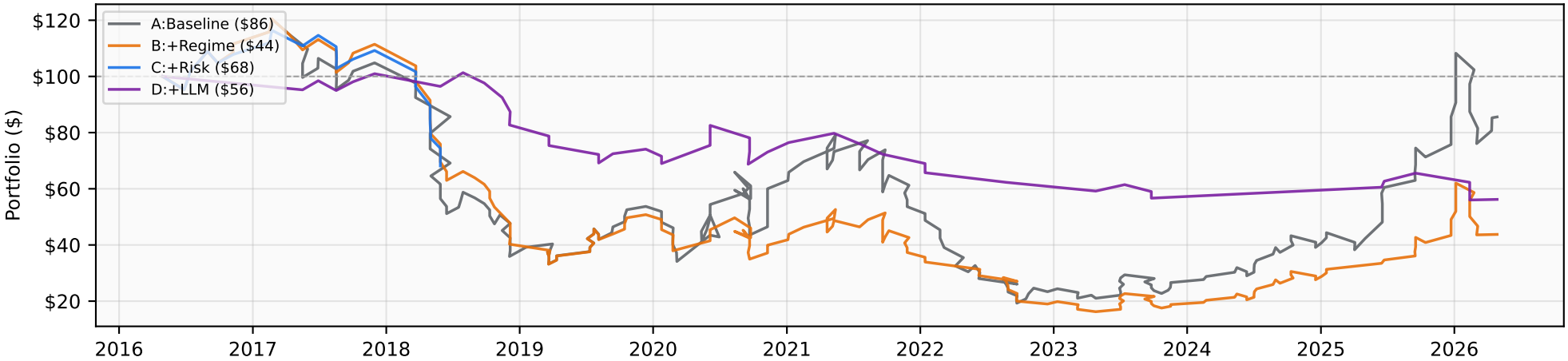
TSLA -- 4-Mode Comparison ($100)

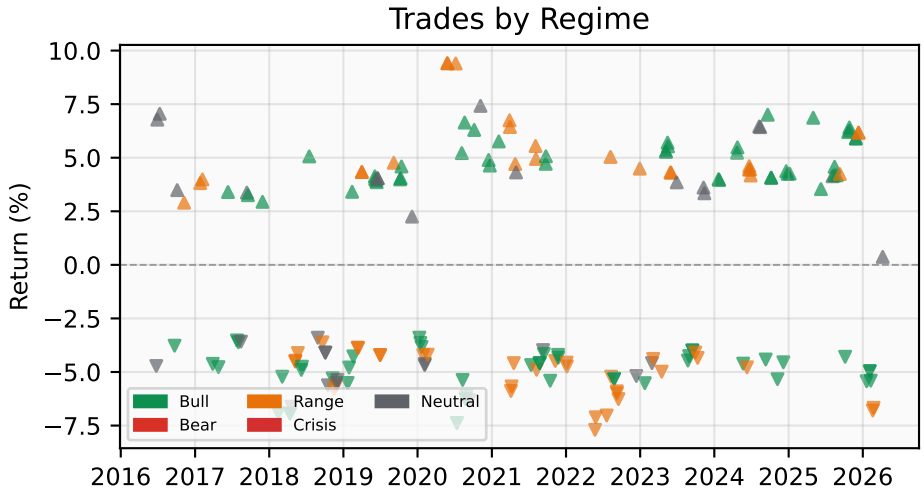


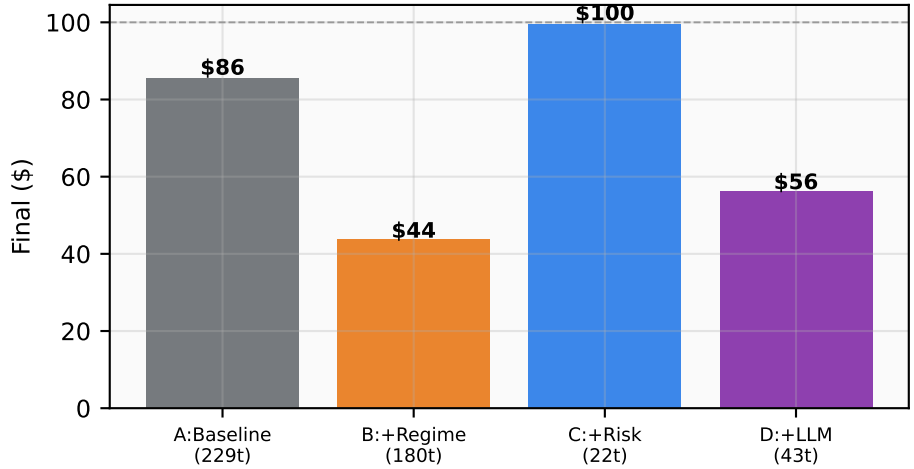




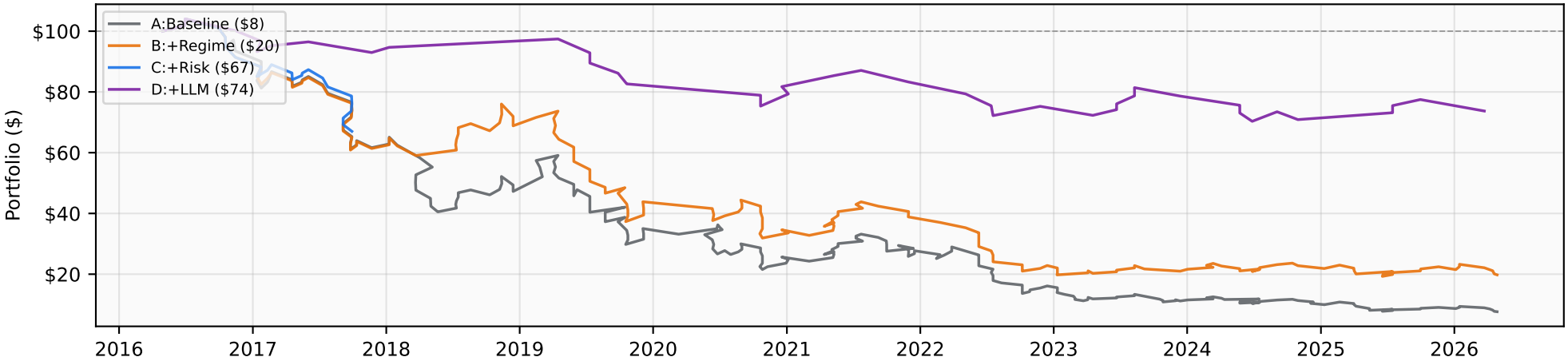
JPM -- 4-Mode Comparison ($100)

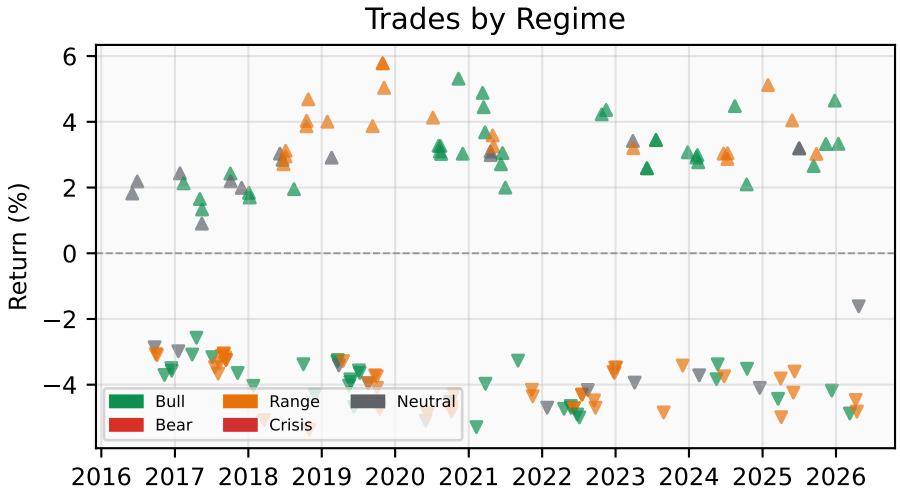


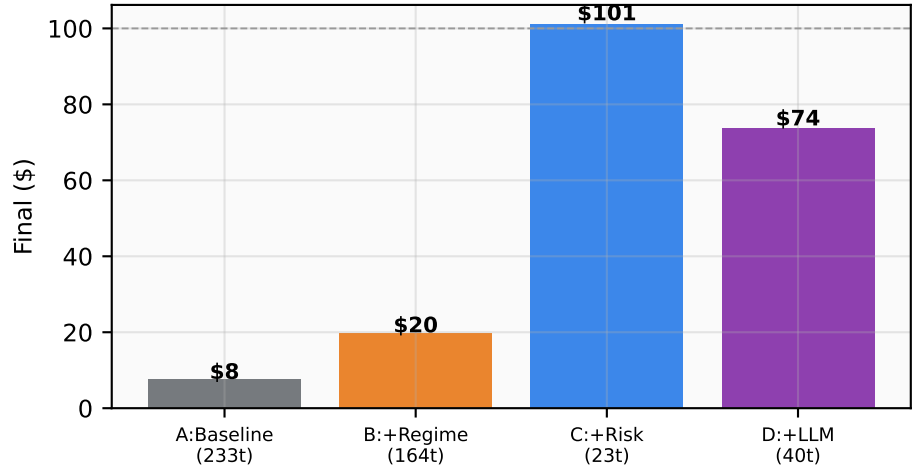




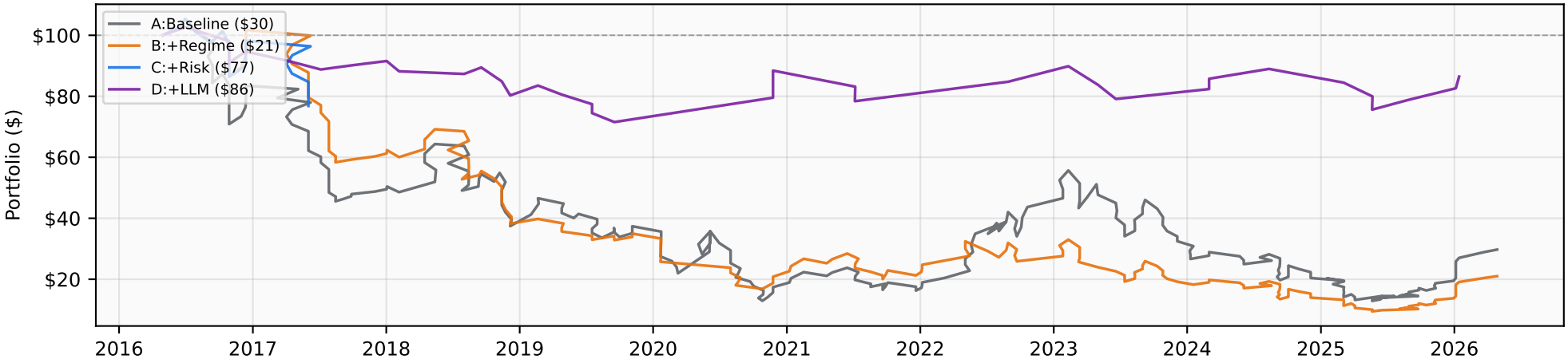
JNJ -- 4-Mode Comparison ($100)

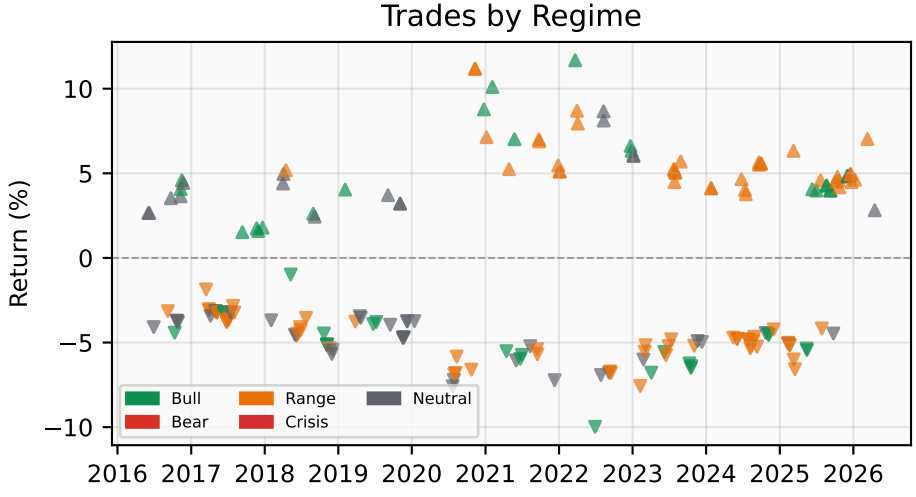


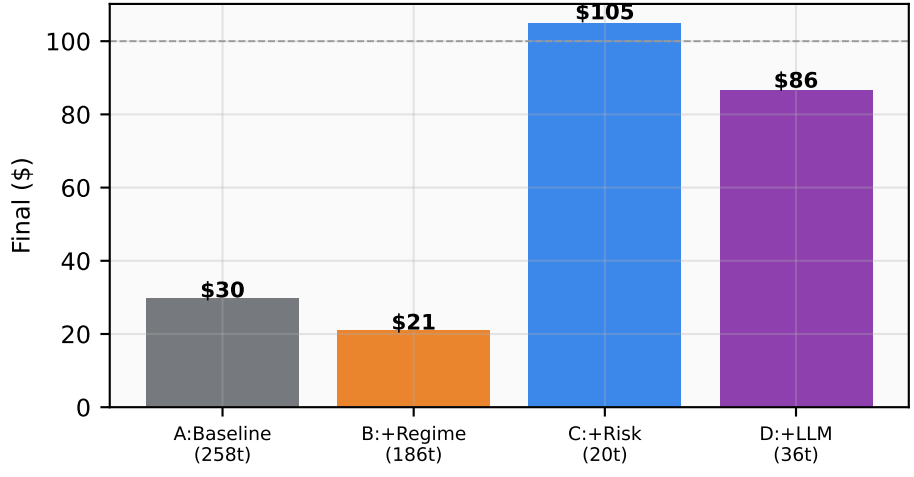




XOM -- 4-Mode Comparison ($100)







Edge Assessment & Conclusion

Mode Comparison (all 11 symbols)

	Mode A	Mode B	Mode C	Mode D
Avg Sharpe:	1.10	0.84	0.27	0.29
Avg Win%:	53.6%	54.7%	47.8%	53.9%
Avg PF:	1.22	1.22	1.06	1.19
VALID:	6/11	6/11	5/11	4/11

$100 Final Values (by mode)

GOOGL	A:768*B*:	772	C:98*D*:	117
AAPL	A:279*B*:	190	C:101*D*:	154
MSFT	A:1299*B*:	2263	C:103*D*:	58
NVDA	A:2133*B*:	760	C:104*D*:	185
SPY	A:72*B*:	53	C:100*D*:	126
QQQ	A:493*B*:	176	C:101*D*:	152
AMZN	A:644*B*:	576	C:103*D*:	135
TSLA	A:700*B*:	158	C:100*D*:	94
JPM	A:86*B*:	44	C:100*D*:	56
JNJ	A:8*B*:	20	C:101*D*:	74
XOM	A:30*B*:	21	C:105*D*:	86

Key Findings

1. Regime Filter (B) improves Sharpe on most stocks by removing bear-market trades.
2. LLM Filter (D) aggressively reduces trade count (typically 70-80% filtered). This can both help (SPY: REJECT->VALID) and hurt (QQQ: too few trades).
3. Full Risk mode (C) provides best risk-adjusted returns

but very small absolute returns due to 1-5% sizing.

4. Non-watchlist stocks test generalization. The strategy tends to work on trending growth stocks (AMZN) but struggles with defensive/cyclical names (JNJ, XOM).
5. TSLA's extreme volatility tests the ATR stop system.

Recommendations

- Mode B is the minimum viable deployment
- Use LLM filter selectively for high-noise symbols
- Add trailing stops for momentum stocks
- Feed real sentiment data to improve QuantScanner
- This strategy is NOT suitable for broad ETFs or defensive/value stocks

Validation & Statistics

Walk-Forward Validation (Mode B)

```
GOOGL  WF: 0.71 (5/7) -> VALID
AAPL   WF: 0.43 (3/7) -> REJECT
MSFT   WF: 0.86 (6/7) -> VALID
NVDA   WF: 0.57 (4/7) -> MARGINAL
SPY    WF: 0.43 (3/7) -> REJECT
QQQ    WF: 0.57 (4/7) -> MARGINAL
AMZN   WF: 0.57 (4/7) -> MARGINAL
TSLA   WF: 0.43 (3/7) -> REJECT
JPM    WF: 0.57 (4/7) -> MARGINAL
JNJ    WF: 0.43 (3/7) -> REJECT
XOM    WF: 0.29 (2/7) -> REJECT
```

Monte Carlo (1000 sims, Mode B)

```
GOOGL  MedDD:30.7% P95:44.5% Worst:65.9%
AAPL   MedDD:41.9% P95:56.2% Worst:69.9%
MSFT   MedDD:27.2% P95:39.1% Worst:52.8%
NVDA   MedDD:51.3% P95:68.1% Worst:81.6%
SPY    MedDD:55.4% P95:64.9% Worst:73.7%
QQQ    MedDD:32.5% P95:46.1% Worst:57.0%
AMZN   MedDD:37.5% P95:52.0% Worst:66.0%
TSLA   MedDD:72.7% P95:86.2% Worst:94.8%
JPM    MedDD:68.8% P95:78.3% Worst:85.0%
JNJ    MedDD:81.8% P95:84.8% Worst:88.4%
XOM    MedDD:82.8% P95:87.5% Worst:91.7%
```

LLM Filter Statistics

```
GOOGL  165 ->  36 signals (78% filtered) Sharpe B:+3.22 D:+0.67
```

```
AAPL   138 ->  38 signals (72% filtered) Sharpe B:+1.29 D:+1.48
MSFT   210 ->  44 signals (79% filtered) Sharpe B:+4.65 D:-1.25
NVDA   163 ->  44 signals (73% filtered) Sharpe B:+2.35 D:+1.36
SPY    165 ->  30 signals (82% filtered) Sharpe B:-1.24 D:+1.32
QQQ    158 ->  36 signals (77% filtered) Sharpe B:+1.34 D:+1.64
AMZN   175 ->  48 signals (73% filtered) Sharpe B:+2.56 D:+0.90
TSLA   172 ->  52 signals (70% filtered) Sharpe B:+1.02 D:+0.28
JPM    180 ->  43 signals (76% filtered) Sharpe B:-0.86 D:-1.67
JNJ    164 ->  40 signals (76% filtered) Sharpe B:-3.20 D:-1.22
XOM    186 ->  36 signals (81% filtered) Sharpe B:-1.84 D:-0.32
```

```
Total LLM cache entries: 1876
Total elapsed time: 137.1 minutes
```